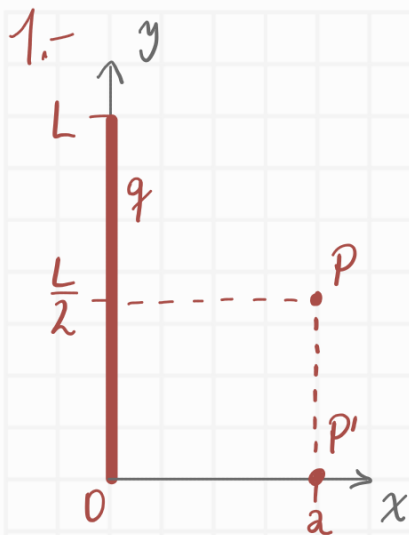




a) $\vec{r} = a\hat{i} + \frac{L}{2}\hat{j}$
 $\vec{r}' = y\hat{j}, \forall y \in [0, L]$

$$dq = \lambda dy$$

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{\vec{r} - \vec{r}'}{\|\vec{r} - \vec{r}'\|^3} dq$$

$$\vec{r} - \vec{r}' = a\hat{i} + (\frac{L}{2} - y)\hat{j}$$

$$\|\vec{r} - \vec{r}'\| = \sqrt{a^2 + (\frac{L}{2} - y)^2}$$

$$= \frac{1}{4\pi\epsilon_0} \int_0^L \frac{\lambda(a\hat{i} + (\frac{L}{2} - y)\hat{j})}{(a^2 + (\frac{L}{2} - y)^2)^{3/2}} dy$$

Por simetría, $\vec{E}_y = 0$ (demostrar por qué)

$$\vec{E}_x = \frac{\lambda a}{4\pi\epsilon_0} \int_0^L \frac{1}{(a^2 + (\frac{L}{2} - y)^2)^{3/2}} dy$$

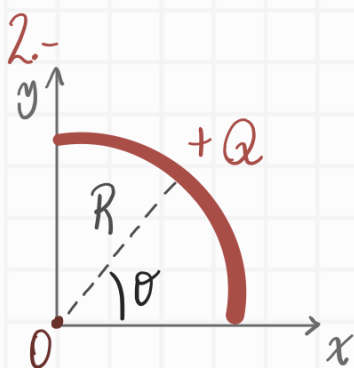
$$u = \frac{L}{2} - y \quad u(0) = \frac{L}{2}$$

$$du = -dy \quad u(L) = -\frac{L}{2}$$

$$= \frac{\lambda a}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{1}{(a^2 + u^2)^{3/2}} du \quad (\text{Continuar!!!})$$

b) $\vec{r} = a\hat{i}$
 $\vec{r} - \vec{r}' = a\hat{i} - y\hat{j}$
 $\|\vec{r} - \vec{r}'\| = \sqrt{a^2 + y^2}$

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{a\hat{i} - y\hat{j}}{(a^2 + y^2)^{3/2}} dy$$



$$\vec{r} = 0$$

$$\vec{r}' = R(\cos\theta\hat{i} + \sin\theta\hat{j})$$

$$\left. \begin{array}{l} dq = \lambda dl \\ l = \theta R \end{array} \right\} dq = \lambda R d\theta \quad \forall \theta \in [0, \frac{\pi}{2}]$$

$$\vec{r} - \vec{r}' = -R(\cos\theta\hat{i} + \sin\theta\hat{j})$$

$$\|\vec{r} - \vec{r}'\| = \sqrt{R^2\cos^2\theta + R^2\sin^2\theta} = R$$

$$\vec{E} = \frac{\lambda}{4\pi\epsilon_0} \int_0^{\pi/2} \frac{-\cos\theta\hat{i} - \sin\theta\hat{j}}{R^2} d\theta$$

$\swarrow$ $\searrow$
 x y
 (Continuar!!!)